SYNACKTIV

TRAININGS



2026.04

Presentation

Synacktiv is committed to sharing of its experience in cybersecurity, acquired over the years, by providing intercompany training. Combining theoretical teaching and practical work, our training courses have been designed to offer a unique and enriching learning experience, and are mainly aimed at **information security professionals**: pentesters, reverse-engineering experts, SOC analysts, CSIRT analysts, system administrators, security architects, developers, etc.

Each session is led by **two experienced trainers** who will ensure optimal understanding while providing concrete feedback. All the material necessary for the realization of the practical work will be provided to the students and each will have an individual environment in order to ensure an **immersive learning experience**. The course materials will be transmitted in PDF format, allowing participants to consult them at any time and use them as a reference.

The trainings take place in our Parisian offices, in a professional and comfortable environment which will promote the concentration of learners. **Lunches and drinks are included**, as well as a restaurant meal on the last day of training.



- 2 experienced trainers
- 7 to 12 participants
- Minimum 50% practice
- Practical work in individual labs
- Equipment provided (laptops)
- Course materials provided
- In our offices 5 bd Montmartre, Paris 75002
- Meals and drinks included

Pentest

Offensive Security Fundamentals

Obtain the skills needed to understand the main phases of an intrusion. Reconnaissance, web applications, Linux and Windows systems, post-exploitation steps, this training provides an essential base for any security professional.

5 days | Junior

Active Directory Intrusion Tactics: Entry Level

Discover the fundamentals of security in Active Directory environments through this offensive training. From anonymous access to the complete compromise of infrastructures, become autonomous in intruding corporate networks.

5 days | Intermediate

DevOps & Linux Breach Tactics

Master intrusion techniques on DevOps and Linux infrastructures through this offensive training. From compromising microservices and injecting CI/CD pipelines to poisoning artifacts, gain, become autonomous in intruding corporate networks.

5 days | Advanced

Active Directory Intrusion Tactics: Advanced Level

Deepen your intrusion skills in Active Directory environments with this advanced level training. Learn advanced exploitation techniques and master the compromise of complex corporate networks.

5 days | Advanced

Cloud Intrusion Tactics

Learn about modern network compromise with this pffensive cloud infrastructure training. GCP, AWS, Azure and Kubernetes, discover the characteristic mechanisms of these recent technologies and the specificities of each provider, with the posture of an attacker.

5 days | Intermediate

Azure Intrusion Tactics

Refine your intrusion skills on Azure with this advanced training. Dive into Entra ID, M365 services, Azure resources, CI/CD with Azure DevOps, Intune management, and hybrid environments through subtle and realistic exploitation scenarios.

5 days | Advanced

Pentest

Kubernetes Intrusion Tactics

Get introduced to compromising containerized platforms with this course dedicated to Kubernetes. Understand Kubernetes-specific authentication, authorization, networking, and escape mechanisms by adopting an attacker's mindset to identify vulnerabilities and remediation paths.

2 days | Junior

AWS Intrusion Tactics

Get started with compromising AWS environments with this course focused on offensive cloud techniques. Understand IAM, EC2 metadata exploitation, S3 abuse, Lambda techniques, and lateral movement, adopting an attacker's mindset to identify vulnerabilities and remediation paths.

2 days | Junior

Attacking Web Applications

Learn about modern web application security mechanisms and advanced exploit methods to circumvent them. PHP, Java, Python and Perl, master the compromise of complex web applications.

5 days | Intermediate

Practical Web 0-Day Hunting

Obtain the skills needed to search for Java, PHP and .NET web vulnerabilities. Study of static and dynamic analysis frameworks and tools, this training allows pentesters and developers to optimize their search for vulnerabilities in white box.

5 days | Advanced

Offensive CI/CD

Master CI/CD exploitation techniques on GitHub and GitLab with this intensive offensive training. From pipeline injection to runner hijacking and secret extraction, learn to execute complex kill chains and compromise the backbone of modern software development.

2 days | Intermediate

Attacking Android Applications

Discover methodologies and techniques for analyzing Android applications. Architecture, entry points, static and dynamic analyses, master the pentesting methodology of the Android environment.

2 days | Junior

Pentest

Password Cracking

Study methods for optimizing password cracking with John and Hashcat tools. Mutation rules, masks, prince and siga attacks, become a real password expert.

1 day | Junior

Reverse Engineering

Embedded Linux Exploitation

This 5-day, hands-on training provides a comprehensive methodology for the security assessment of embedded Linux systems. Working with a real-world device, participants will progress through the complete vulnerability research workflow, from hardware analysis to exploitation.

5 days | Junior

Hardware Intrusion

Learn to tame a PCB: recognize components, identify testpads, infer and then interact with protocols (UART, JTAG/SWD, SDIO, SPI). Use the active/passive tools and materials (analyzer logic, FT2232H, JTAGulator, OpenOCD).

5 days | Intermediate

Android for Security Engineers

Discover the internals of the Android operating system and its security mechanisms with the help of practical exercises.

5 days | Intermediate

iOS for Security Engineers

Discover the internals of the iOS operating system and its security mechanisms with the help of practical exercises.

5 days | Intermediate

Advanced IDA

Familiarize yourself with the advanced features of IDA, its API and its ecosystem. Learn how to develop scripts and plugins to extend its functionalities.

5 days | Advanced

DMA Attacks

Master hardware attacks and direct memory access (DMA) exploitation. From manipulating capture equipment to on-the-fly modification with PCILeech and RAM forensic analysis, learn how to physically compromise workstations.

4 days | Intermediate

Forensics

Windows Forensics

Master the digital investigation of Windows 10 and 11 systems by learning to identify and characterize the associated malicious actions, both in the context of a security incident and a search for compromise (removal of doubt, hunting).

5 days | Junior

Linux Forensics

Master the digital investigation of Linux systems by learning to identify and characterize the associated malicious actions, both in the context of a security incident and a search for compromise (removal of doubt, hunting).

5 days | Junior

Mobile Forensics

Discover the digital investigation of Android and iOS mobile operating systems by studying data acquisition techniques, the discovery of malicious applications and phone artifacts.

5 days | Junior

Kubernetes Forensics

Master digital forensics in Kubernetes environments. Understand the inner workings of cluster components to respond effectively to compromises. Identify threats and maintain order within the cluster.

3 days | Intermediate

Malware Analysis

Discover the analysis of malicious code in the context of a security incident through various situations and real cases of attackers' operating methods.

3 days | Intermediate

Advanced Malware Analysis

Reverse-engineer threats from various environments and discover methods to tackle obfuscated and evasive malware through scripting.

3 days | Advanced

Forensics

Cloud Forensics in AWS

Take a deep dive into AWS to understand how the logging system works. Learn how to detect main attacks on these environments and specific use cases for reconstructing threat actor activity.

3 days | Junior

Cloud Forensics in Azure

Take a deep dive into Microsoft Azure, Entra ID and M365 to understand how the logging system works. Learn how to detect main attacks on these environments and specific use cases for reconstructing threat actor activity.

3 days | Junior

Ransomware Investigation

Understand first aid and emergency measures for a ransomware incident. Continue your understanding of the situation with investigation methods adapted to this threat and start remediation.

3 days | Intermediate

Active Directory: Hardening & Post-Compromise Recovery

Secure the core of your IT system by mastering Active Directory hardening, monitoring, and remediation. From tiering deployment to post-intrusion trust recovery, learn how to block attackers and rebuild a healthy environment.

5 days | Intermediate

Data Breach: Investigations & Compliance

Master crisis management during a data breach by combining emergency technical investigations (quick wins) and legal compliance. Learn to classify the incident, manage communication, and comply with your legal obligations in various compromise scenarios.

1 day | Junior

Development

Agentic AI Red Teaming

Master the development of autonomous architectures by designing a Red Team agent capable of reasoning and acting. Learn to orchestrate LLMs and integrate tools via the MCP protocol.

5 days | Intermediate

Network Interception in Rust

Design a high-performance Man-In-The-Middle interception tool from scratch in Rust to master local network attacks (ARP/DHCP) and the manipulation of encrypted application flows (HTTP/TLS).

3 jours | Intermediate

Advanced Rust

Master the advanced concepts of the Rust language (multi-threading, asynchronicity, typing) by iteratively developing a high-performance file scanner, from end to end, up to its control via a web interface.

5 days | Advanced

Infrastructure

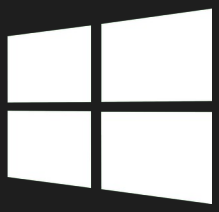
Linux Hardening

Learn how to drastically reduce the attack surface of your Linux systems using native defense mechanisms. From application isolation with AppArmor and Podman to granular network filtering via Nftables, build robust environments capable of preventing any compromise.

4 days | Intermediate

Offensive Security Fundamentals

5 days | Junior level



Description

Performing intrusion tests allows a realistic simulation of defense mechanisms and therefore represents a key step in securing information systems. This introductory pentest training aims to provide an in-depth understanding of security auditing by addressing the different stages of an intrusion.

During these five days of training, participants will be exposed to four course modules covering reconnaissance, web applications, Linux and Windows systems, and post-exploitation techniques. Each module will be illustrated by guided practical work to apply the theoretical notions taught. Finally, the training will conclude with a realistic scenario on a corporate network.

- 5 days (35 hours)
- 4 course modules covering the main steps of a penetration test
- Reconnaissance, web applications, Linux, Windows, post-exploitation
- 20 exercises
- 1 guided intrusion on a complete corporate environment (10 machines)

Audience and prerequisites

This training has been designed for people with no prior experience in penetration testing. It is mainly aimed at beginner pentesters, system administrators, security architects and developers, but also at any technical profile wishing to enrich their professional career with a security component.

- Beginner pentesters
- System administrators
- Security architects
- Developers

Basic knowledge of the Unix environment and web languages is recommended.

Content

Day 1

Introduction to discovery methods: DNS and HTTP enumeration, service scans. Overview of the main intrusion tools: Metasploit, Burp Suite. Web application vulnerabilities: SQL injections, XSS (Cross-Site Scripting), XXE (XML eXternal Entities), SSRF (Service-Side Request Forgery), file upload, deserialization, with various practice exercises.

Day 2

Practice on complex web applications: reconnaissance, exploitation and elevations of privileges until obtaining access to servers. Privilege escalation on Linux systems: fundamentals (identity and access management), reconnaissance and exploitation (permissions, sudo configurations, scheduled tasks, systemd units, kernel), and containerization technologies (Docker, LXC/LXD).

Day 3

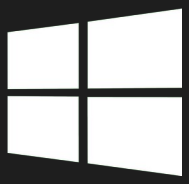
Privilege elevation on Windows systems: fundamentals (identity and access management, secrets management), reconnaissance and exploitation (permissions, service configurations, scheduled tasks, public vulnerabilities). Hands-on practice on servers from non-privileged access.

Days 4 and 5

Post-exploitation steps: extracting secrets (disk, memory), installing backdoors and lateral moves (network bounce, SOCKS proxy, port forward). Hands-on intrusion of a corporate network.

Active Directory Intrusion Tactics: Entry Level

5 days | Intermediate level



Description

For many companies, Active Directory is the heart of identity and access management. Its ubiquity within information systems makes it a prime target for computer attacks, and penetration testing is a key component of its defense against threats.

During this five-day training, you will learn the skills necessary to perform an in-depth Active Directory penetration test. By following the five course modules, students will learn the methodology and techniques used by our experts during an intrusion, from anonymous access to the complete compromise of the environment and the persistence of access within it. To illustrate new concepts, learners will be guided through two comprehensive corporate environments.

- 5 days (35 hours)
- 5 course modules covering all intrusion steps + 1 Azure module
- 2 corporate environments with more than 40 machines and an Azure environment

Audience and prerequisites

This training is suitable for people with notions of offensive security but no prior experience in Active Directory environments. It is aimed primarily at pentesters, system administrators and security architects, but also at any technical profile wishing to enrich their professional career with a security component.

- Pentesters
- System administrators
- Security architects

Notions of offensive security and good network and Unix knowledge are recommended.

Content

Day 1

Theoretical foundations of security mechanisms: administration mechanisms (RPC, SMB, WMI, RDP, WinRM), identity and access management, storage of secrets, network authentication protocols (NTLM, Kerberos), hierarchy and Active Directory trusts. Reconnaissance and exploitation techniques from anonymous access: enumeration, network protocol poisoning, relaying.

Day 2

Reconnaissance on the domain from non-privileged access: objects extraction (users, groups, machines, GPOs) and mapping with BloodHound. Local privilege escalation: enumeration and exploitation (local services, scheduled tasks, ACLs, public vulnerabilities), UAC bypass techniques.

Day 3

Escalation of privileges within a domain: secrets extraction (registry, LSASS, DPAPI), replay of authentication, kerberoasting, abuse of control paths. Bypassing software restrictions: AppLocker, evading restricted desktops (Citrix, RDP Kiosk).

Day 4

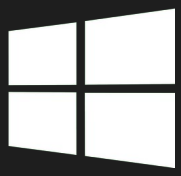
Post-exploitation steps from privileged access on the domain: secrets extraction (NTDS, DPAPI), ticket forgery (silver and golden tickets), manipulating ACLs, persisting within the environment, and erasing traces. Extending the compromise: cross-domain and cross-forest trust relationships, Kerberos delegation abuse.

Day 5

Introduction to Azure: fundamental concepts (terminology, identity and access management), integration with Active Directory (identity synchronization, Single Sign-On mechanisms), recognition and compromise steps from the on-premise environment.

Active Directory Intrusion Tactics: Advanced Level

5 days | Advanced level



Description

For many companies, Active Directory is the heart of identity and access management. Its ubiquity within information systems makes it a prime target for computer attacks, and penetration testing is a key component of its defense against threats.

During this five-day training, you will deepen your intrusion skills in an Active Directory environment. Guided by our experts, study advanced techniques of reconnaissance, lateral movements, elevation of privileges, extraction of secrets and persistence. To illustrate new concepts, the learners will be put in situation on two complete company environments derived from real-world scenarios.

- 5 days (35 hours)
- 5 course modules covering all intrusion steps
- 2 corporate environments with 30+ machines and specific services such as ADCS and SCCM

Audience and prerequisites

This training is designed for experienced technical professionals with significant prior experience in Active Directory environments. It is the logical continuation of the Active Directory Intrusion Tactics: Entry Level course. Target audience:

- Penetration testers / red teamers
- System administrators
- Security architects

To fully benefit from this training, participants should have prior experience in the following areas:

- Proficient use of the Impacket suite, NetExec, and BloodHound.
- Mastery of basic attacks: LLMNR/NBNS poisoning, Kerberoasting, and credential replay.
- Extraction of secrets related to Windows environments (SAM database, LSASS processes).
- Experience with network hopping and tunneling (SOCKS/TUN) for navigating between different networks.

A good understanding of Windows and UNIX administration, as well as NTLM and Kerberos protocols, is recommended for the advanced modules.

Content

Day 1

Fundamentals: Active Directory mechanisms, general and specific intrusion principles for these environments. Recognition and first actions from authenticated access: information retrieval methods (ADIDNS, service detection via LDAP and GPO scans) advanced use of BloodHound (Cypher queries).

Day 2

Lateral movements: ADIDNS, WinRM and JEA poisoning, LAPS, gMSA/sMSA secrets extraction, MS-SQL trust abuse, NTLM relaying (dissection, cross-protocol relaying, WebDAV), authentication coercing, Kerberos relaying, cross-forest pivots.

Day 3

Local privilege elevation: access token and impersonation, study of potatoes vulnerabilities. Escalation of privileges on the domain: study and abuse of ACLs, advanced exploitation of Kerberos delegation, ADCS ESC1 to 15, SCCM mechanisms and exploitation primitives, abuse of privileged groups, analysis of public vulnerabilities.

Day 4

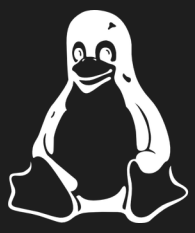
Secrets extraction: LSASS dump methods and tools, token spoofing, registry secrets analysis, DPAPI implementation, KeePass database.

Day 5

Persistence: ADCS (certificates), Kerberos tickets (golden, diamond, sapphire), DSRM, golden gMSA, AdminSDHolder abuse, skeleton key creation, Kerberos delegation, GPO poisoning, DC Shadow.

DevOps & Linux Breach Tactics

5 days | Advanced level



Description

DevOps environments heavily rely on Linux to power microservices architectures, hybrid infrastructures, and continuous deployment pipelines. The security of these environments depends on automation and containerization tools, the mastery of which has become a critical challenge for modern attackers.

Over this five-day course, participants will be exposed to five theoretical modules detailing a complete kill chain: from initial access via CI/CD pipeline injection (GitLab, Jenkins) to the abuse of Infra-as-Code tools (Ansible, Terraform) to compromise the infrastructure. An additional module will be dedicated to hardened systems (AppArmor, SELinux). These concepts will be applied throughout the week on two complex corporate networks, inspired by real-world intrusions conducted by our experts.

- 5 days (35 hours)
- 5 course modules following realistic intrusion steps + 1 module on hardened systems
- 2 corporate environments with more than 20 machines including GitLab, Jenkins, Artifactory, AWX, HashiCorp Vault, Guacamole, KeyCloak, and vSphere services
- RedHat-like and Debian-like distributions

Audience and prerequisites

This training is suitable for people with notions of offensive security but no prior experience in the intrusion of corporate DevOps and Linux environments. It is aimed primarily at pentesters, system administrators and security architects, but also at any technical profile wishing to enrich their professional career with a security component.

- Pentesters / red teamers
- System administrators
- Security architects

Solid Unix/Linux skills, networking basics, and an interest in offensive security are recommended.

Content

Day 1

Fundamental concepts: Identity and Access Management (IAM), security mechanisms (extended ACLs, standard and extended attributes, capabilities), networking fundamentals, containerization (namespaces, cgroups, seccomp, Docker and LXC / LXD). Microservices: Docker registry, Portainer, Traefik, Redis, container detection, image analysis, secret hunting, and backdooring. Centralized authentication: Kerberos, OpenLDAP, SSSD and FreeIPA, reconnaissance, enumeration, and tooling.

Day 2

Lateral movement: network pivoting, SOCKS proxy, port forwarding, TUN forwarding, and tooling. CI/CD: Jenkins, pipeline injection and agent compromise, Artifactory, dependency confusion and artifact poisoning. Workstations: KeePass, DBUS Secret Service API, SSH and GPG agents, Firefox and Chrome browsers, keyloggers.

Day 3

Administration: PKI and certificate signing, firewall manipulation (iptables / nftables), derouting (DNAT) and spoofing (SNAT), file systems (NFS, Samba, FUSE), IAM (KeyCloak), bastion (Guacamole), packaging and poisoning (Deb and RPM). Hypervisors (vSphere): architecture, ESXi and vCenter, authentication methods, post-exploitation techniques (network capture and bypass, optimized exfiltration), golden SAML.

Day 4

GitLab: IAM study, runner architecture and implementation, reconnaissance, initial access via dumps and project analysis (TruffleHog, noseyparker), privilege escalation via pipeline injection, secrets dumping. Infra-as-Code: Ansible, Terraform, AWX, and automated deployments. HashiCorp Vault: architecture, authentication, access policies, and usage.

Day 5

Advanced exploitation: process management (sessions, groups and lifecycle), TTY internals and injections, PAM stack analysis, and backdooring. Compromising hardened systems: implementation and configuration of AppArmor and SELinux LSMs, hardening analysis and bypass techniques.

Cloud Intrusion Tactics

5 days | Intermediate level



Description

Cloud technologies are gradually being integrated into the information system of companies. They provide many security mechanisms that are sometimes difficult to understand and force attackers to rethink their methods of intrusion.

During this five-day course, participants will be exposed to the concepts of the three major cloud providers: GCP (Google), AWS (Amazon), and Azure (Microsoft). After having studied the fundamentals they share, their implementation specificities will be detailed and illustrated through complete environments allowing to learn about cloud intrusion techniques. An additional module will also be dedicated to Kubernetes infrastructures.

- 5 days (35 hours) customizable
- 3 course modules on GCP, AWS and Azure + 1 module dedicated to Kubernetes
- 4 complete and individual environments

Audience and prerequisites

This training is suitable for people with notions of offensive security, but no prior experience in cloud environments. It is aimed primarily at pentesters, system administrators, security architects and developers, but also at any technical profile wishing to enrich their professional career with a security component.

- Pentesters / red teamers
- System administrators
- Security architects
- Developers

Good network and Unix knowledge and notions of web intrusion are recommended.

Content

Day 1

Fundamentals: cloud terminology, infrastructure services, network topology, identity and access management, authentication mechanisms (OAuth), reminders of Linux security mechanisms (namespaces, cgroups, seccomp, LSM), OSINT.

Day 2

Google Cloud Platform: architecture (organization, folders, projects, resources, regions, and zones), IAM (permissions, roles, principals, and policies), authentication (OAuth 2.0, JWT), using the gcloud CLI, service discovery methods, abuse of rights on buckets, App Engine and instance implementations (metadata abuse), elevation of IAM privileges, network reconnaissance (VPC, firewall, VPN, peerings), post-exploitation (delegation on the domain, bounce on Workspace), analysis events.

Day 3

Amazon Web Services: architecture (organization, accounts), IAM (identity types, role assumption, policies), aws CLI usage, service discovery methods, unauthenticated identity enumeration, S3 bucket rights abuse, EC2 (metadata, lateral movements and poisoning of SSM agents), Lambdas (runtime API, persistence, data exfiltration), Cognito (user and identity pools) IAM privilege escalation, network reconnaissance (VPC, network ACL, security groups) , persistence (modification of IAM policies, role chain juggling).

Day 4

Azure: architecture (tenants, management groups, subscriptions), Entra ID (identity types, access management, Entra vs Azure roles), synchronization in hybrid environment (PHS, PTA, ADFS), unauthenticated discovery, use of azure CLI and Az module, authenticated discovery (ROADrecon, AzureHound), blob storage implementation, key vault, virtual machines, lateral movements (Vnet, bastions).

Day 5

Kubernetes: architecture (containers, pods, nodes, internal services), recognition, authentication (password, certificates, tokens) and authorizations (node, ABAC, RBAC, WebHook), kubectl CLI usage, pod templates and controllers, escapes (namespaces, PSP, PSA), network concepts (ingress, pod to pod, CNI, policies).

Azure Intrusion Tactics

5 days | Advanced level



Description

Azure is currently an undisputed leader in cloud computing, ubiquitous in enterprise infrastructures due to its close integration with on-premise environments, particularly through Active Directory. This massive adoption makes it a prime target for attackers, who must understand its security specifics in order to conduct effective penetration tests.

During this five-day training, participants will deepen their offensive skills on Azure. After an introduction to fundamental security principles, the focus will shift to key services such as Entra ID, Microsoft 365 suite, Azure resources, CI/CD with Azure DevOps, Intune management, and hybrid environments. Realistic and subtle exploitation scenarios will enable participants to acquire the necessary techniques to compromise these infrastructures, while adopting a stealthy and targeted approach.

- 5 days (35 hours)
- 6 course modules covering a complete intrusion of Azure services
- 1 realistic, comprehensive, and individualized environment

Audience and prerequisites

This training is suitable for people with notions of offensive security, but no prior experience in Azure environments. It is aimed primarily at pentesters, system administrators, security architects and developers, but also at any technical profile wishing to enrich their professional career with a security component.

- Pentesters / red teamers
- System administrators
- Security architects
- Developers

Good network and Unix knowledge and notions of web intrusion are recommended.

Content

Day 1

Fundamentals: tenant architecture, portals, tooling. Entra ID: identities and roles, OAuth 2.0 framework and Microsoft implementation, access tokens, application concepts, Microsoft Graph API, security (conditional access policies, PIM, detections), authentication methods, MFA and bypass opportunities, event logging, initial access and discovery (AzureHound, ROADTools, Microsoft Graph CLI).

Day 2

Microsoft 365: analysis of the main services of the productivity suite (Teams, Outlook / Exchange, SharePoint / OneDrive, OneNote / Word / Excel), specific APIs and their offensive usage, access management through Microsoft Graph vs. dedicated APIs, event logging.

Day 3

Azure resources: architecture, reconnaissance and discovery (AzureHound / BloodHound), ARM API, az CLI and portals, virtual machines / VDIs, containers and registries, key vaults, app services, network (firewalling and interconnections) and storage, lateral movements and post-exploitation, study of event logging. Azure DevOps: architecture, CI/CD concepts, access control.

Day 4

Azure DevOps: agents implementation, pipeline injections, privileges escalation and post-exploitation (persistence and secrets extraction). Intune: relations with Entra ID, enrollment process, access management through Intune vs. Entra ID, services implementation on managed devices, post-exploitation (scripts and applications deployment), stealth tooling.

Day 5

Hybrid environments: synchronization methods (PHS, PTA, federation), bi-lateral movements between Active Directory and Azure, AZURESSO, ADFS, implementation of Entra Connect/Cloud Sync agents, cookie theft.

Kubernetes Intrusion Tactics

2 days | Junior level



Description

Kubernetes is now an omnipresent technology in modern infrastructures. Used to automate deployment, scaling and management of applications, it has become a cornerstone of DevOps and cloud-native environments. Its complexity and the diversity of its components nevertheless create a rich and often poorly understood attack surface, exposing organizations to new types of vulnerabilities and compromises.

During this two-day course, participants will discover Kubernetes internals and learn how to exploit its weaknesses from an offensive perspective. From component enumeration to container escapes, including privilege escalation and lateral movement, the course combines theory and hands-on practice on a vulnerable cluster. It is aimed at auditors, red teamers and security teams who want to understand the concrete risks related to Kubernetes and strengthen the security of their environments.

- 2 days (14 hours)
- Full course module and individual practice environment
- 30% theory / 70% practice

Audience and prerequisites

This course is suitable for participants with offensive security fundamentals but no prior Kubernetes experience. It is primarily intended for pentesters, system administrators, security architects and developers, and is also appropriate for any technical profile seeking to add a security-focused specialization to their career.

- Pentesters / red teamers
- System administrators
- Security architects
- Developers

Good network and Unix knowledge are recommended.

Content

Day 1

Architecture and attack surface: container fundamentals (Docker, OCI), Kubernetes components (API Server, etcd, Scheduler, Controller Manager, kubelet), typical cluster topology (control plane, worker nodes), internal and exposed services (kube-proxy, dashboard, external APIs), networking concepts (ingress, pod-to-pod, CNI, policies). Initial interactions: reconnaissance, authentication (passwords, certificates, tokens) and authorization (node, ABAC, RBAC, WebHook), use of the kubectl CLI.

Day 2

Advanced concepts: pod templates and controllers, escapes (namespaces, PSP, PSA), lateral movement. Practical exercises and realistic scenarios: exploitation of a vulnerable cluster.

AWS Intrusion Tactics

2 days | Junior level



Description

AWS is now a cornerstone platform for hosting and scaling enterprise applications and services. Its extensive functionality and diversity of services (compute, storage, serverless, managed services) create a large attack surface and specific exploitation patterns that attackers and auditors must master.

In this AWS-focused course, participants will learn AWS architecture, IAM mechanisms, common access vectors (EC2 metadata, permission abuse, public S3 buckets), as well as exploitation of serverless services (Lambda) and network mechanisms (VPC, peering, security groups). The course covers reconnaissance, initial access, privilege escalation, lateral movement, persistence and exfiltration, using a low-footprint approach to preserve stealth. Hands-on practical labs on AWS environments, public tooling, case studies and guided exercises, aimed at pentesters, auditors and security teams seeking to strengthen offensive assessment capabilities on AWS.

- 2 days (14 hours)
- Full course module and individual practice environment
- 30% theory / 70% practice

Audience and prerequisites

This course is suitable for participants with offensive security fundamentals but no prior AWS experience. It is primarily intended for pentesters, system administrators, security architects and developers, and is also appropriate for any technical profile seeking to add a security-focused specialization to their career.

- Pentesters / red teamers
- System administrators
- Security architects
- Developers

Good network and Unix knowledge and notions of web intrusion are recommended.

Content

Day 1

Fundamentals, reconnaissance and initial access: architecture (organization, accounts), IAM (identity types, role assumption, policies), use of the AWS CLI, service discovery techniques, unauthenticated identity enumeration. Exploitation and lateral movement: S3 permission abuse, EC2 (metadata, lateral movement and SSM agent poisoning), Cognito (user and identity pools) and IAM privilege escalation.

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Day 2

Advanced concepts: Lambdas (runtime API, persistence, data exfiltration), network reconnaissance (VPC, network ACLs, security groups), persistence (policy modification, role chaining), log analysis and detection (CloudTrail, CloudWatch, GuardDuty). Practical exercises and realistic scenarios: exploitation of a vulnerable environment.

Attacking Web Applications

5 days | Intermediate level



Description

Web applications represent a large part of the attack surface exposed on the Internet. As technology evolves, new vulnerabilities and exploitation methods continue to emerge, making the intrusion steps more complex.

During this five-day training, participants will study the functioning of the security mechanisms implemented in recent web applications. The various exercises resulting from the feedback of our experts will allow them to refine their intrusion methods for the exploitation of complex vulnerabilities. Finally, learners will be able to understand the specificities of Java, PHP, Python and ASP.NET languages and frameworks, using dedicated modules.

- 5 days (35 hours) customizable
- 9 course modules for Java, PHP, Python and ASP.NET
- Over 30 hands-on exercises

Audience and prerequisites

This training is suitable for people with prior experience in web intrusion techniques. It is mainly intended for pentesters and developers.

- Pentesters
- Developers

Good networking and Unix knowledge is also recommended.

Content

Day 1

BurpSuite: advanced usage, limitations, shortcuts and automation mechanisms, extensions (AuthMatrix, Hackvertor, ActiveScan++). Reconnaissance: DNS enumeration, vhosts, fuzzing, web component identification.

Day 2

Fundamental security mechanisms: authentication (OAuth, JWT, SAML), session management (cookies, tokens, viewstates), password reset, access control, user input management. Advanced exploitation: XXE, SSRF, injections, SSTI, prototype pollution, cryptographic attacks, GraphQL, specifics of cloud environments.

Day 3

Java: recognition and identification of frameworks (extensions, endpoints, headers, administrative interfaces), exploitation of specific vulnerabilities (XXE, HQL injections, deserialization, expression languages, JNDI, path traversals).

Day 4

PHP: recognition and identification of frameworks (endpoints, errors, headers), security functions (session management, sanitization), exploitation of specific vulnerabilities (type juggling, stream wrappers and filters, deserialization and design of complex POP chains, XXE), post-exploitation (fileless execution, disable_functions bypasses).

Day 5

Python Django: attack surface exposure (debug mode, cookie signing, DTL and Jinja2 template injection). ASP.NET: fundamentals, recognition, exploiting specific behaviors (deserialization, ViewState, Web.config, SSTI (Razor), XXE).

Practical Web 0-Day Hunting

5 days | Advanced level



Description

The complexity of modern web applications requires a strong understanding of the native mechanisms of the languages used. Source code analysis methods make it possible to optimize the search for vulnerabilities during an intrusion.

During this five-day course, you will acquire the skills necessary to identify complex vulnerabilities within the source code of PHP, Java and .NET applications. Based on many practical cases on popular frameworks such as Spring or Symfony, participants will learn how to optimize their research using static and dynamic analysis tools.

- 5 days (35 hours) customizable
- 3 course modules covering the specifics of PHP, Java and .NET
- Practical cases on 1-day vulnerabilities
  - Patch diffing
  - Tinted analysis (CodeQL / Semgrep)
  - Instrumentation and debugging
  - Study of main frameworks (Symfony, Spring, etc.)
  - 1-days analysis: vulnerability discovery and exploit creation

Audience and prerequisites

This training is suitable for people with good knowledge of web technologies and associated vulnerabilities. It is mainly intended for pentesters and developers wishing to improve their research method.

- Pentesters
- Developers

A good knowledge of web application security (OWASP Top 10) is required. Basic understanding of networking and Unix is also recommended.

Content

Day 1

Methodology: top-down, bottom-up and hybrid approaches, static and dynamic analysis, tooling, analysis of the application architecture and its environment. PHP: reminders, functioning of the language, security mechanisms, traps, study of known frameworks and analysis of their defense mechanisms and functionalities leading to vulnerabilities, setting up the analysis environment (IDE, Xdebug, PHP configuration, Semgrep).

Day 2

PHP: classic vulnerabilities and exploitation specificities related to PHP, functions leading to vulnerabilities and research methodology. The following topics will be studied: SQL injections, code executions, type juggling, deserialization, wrappers and filters.

Day 3

Java: study of classic Java applications, structure of an application (Class components, JAR, JSP, configurations), formats (WAR, EAR), web.xml configuration (URI mapping, filters, hooks, security constraints), application of top-down and bottom-up approaches, tooling. Specificity of web servers: Tomcat, Jetty, WebLogic, Glassfish, WildFly. Instrumentation and analysis of Java code: setting up a code audit environment, use of an IDE, debugging, instrumentation, decompilation (jd-gui, procyon), tinted analysis using CodeQL, from simple execution of the tool to writing specific queries.

Day 4

Java Spring: dependency injections, beans, controllers, mappings and annotations. Classic vulnerability analyses and exploitation specificities related to Java: LFI, IDOR, XXE and deserialization. Mechanisms implemented by Java to prevent deserialization vulnerabilities (JEP 290, JEP 396) as well as their workarounds.

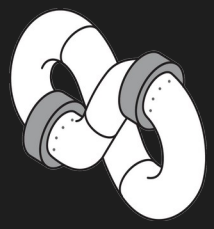
.NET: .NET environment, .NET Framework, .NET Core, ASP.NET and their specificities. IIS Server: operation, configuration, architecture, analysis of a classic deployment and key points to audit. Study of web applications in .NET in order to set up a code audit environment, ways to decompile them and find the key elements for identifying vulnerabilities.

Day 5

.NET deserialization and marshalling: how it works, how it is configured, and how it is used. Understanding existing gadgets to find new ones. .NET Remoting: identifying vulnerable configurations.

Offensive CI/CD

2 days | Intermediate level



Description

Continuous Integration and Continuous Delivery (CI/CD) environments have become the backbone of modern software development. As developers heavily rely on automation to build, test, and deploy applications, these platforms have turned into highly critical targets. Securing these environments and understanding how they can be compromised is now a major challenge for security professionals and modern attackers.

Over this two-day course, participants will be exposed to a complete offensive methodology targeting the two most popular CI/CD solutions: GitHub and GitLab. The training is split into two distinct modules, diving deep into architectural concepts, pipeline abuse, runner hijacking, and post-exploitation. Through realistic corporate environments, attendees will practice complex kill chains, from unauthenticated repository access via pipeline injection to full organizational compromise and secrets extraction.

- 2 days (14 hours)
- 2 intense course modules following realistic CI/CD intrusion steps
- Dedicated corporate environments simulating complex GitHub and GitLab organizations
- Deep dive into real-world vulnerabilities and custom exploitation tools

Audience and prerequisites

This training is suitable for people with notions of offensive security who wish to understand how to attack and secure CI/CD environments. It is aimed primarily at pentesters, DevSecOps engineers, system administrators, and security architects, but also at any technical profile wishing to enrich their professional career with a cloud/CI security component.

- Pentesters / Red Teamers
- DevOps / DevSecOps engineers
- System administrators
- Security architects

Fluent UNIX/Linux skills, solid networking basics, and an interest in offensive security are highly recommended.

Content

Day 1 : GitHub

Architecture & IAM: Core concepts of GitHub organizations, repositories, and identity management (users, agents, and GitHub Apps). Authentication & Authorization: Personal Access Tokens (PAT) and scopes, SSH keys, and the Roles/ACL system (who can create branches, commit, merge, or read logs). Pipelines: GitHub Actions internals, YAML definitions, available modules, and variable stores (managing secrets and the implications of protected branches). Pipelines Exploitation: Injecting malicious code via untrusted inputs, expression injections, dangerous artifacts, and environment manipulation. Exploring supply chain vectors like repo jacking and Dependabot abuse. Runners & OIDC: Hosted vs. self-hosted runners. Runner architecture, registration, and hijacking techniques. OIDC internals, token generation, and the exploitation of weak claim bindings. Tooling & Automation: Utilizing octoscan for static vulnerability analysis and nord-stream for automated CI/CD secret extraction.

Day 2 : GitLab

Architecture & ACLs: GitLab-specific project and group hierarchies, and advanced permission models. Authentication Mechanisms: User/password flows, Personal Access Tokens (PAT) and scopes, SSH keys and external Identity Providers. Pipelines: GitLab CI/CD YAML configuration, specific environment variables, and the security implications of protected variables and branches. Runner Implementation & Hijacking: GitLab agent registration, authentication, filesystem hierarchy, and execution modes (e.g., Shell, Docker, Kubernetes). Techniques to hijack, abuse, and pivot from these runners (persistence, runner identity theft, Docker-in-Docker exploitation). Post-Exploitation: Interacting with the GitLab Ruby console, advanced secret extraction, and accessing repositories directly from the underlying filesystem. Defense & Operations: Analyzing pipeline logs, identifying security alerts, and tracking malicious CI/CD activity.

Attacking Android Applications



2 days | Junior level

Description

Android is one of the most popular mobile operating systems on the market and on which many applications are developed. This ecosystem defines standards for implementation, communication, storage and security mechanisms that are specific to it and that developers must respect.

During this two-day training, participants will discover the specificities of implementing Android applications and will study the methodologies and techniques used to analyze them.

- 2 days (14 hours)
- 2 course modules
- 9 Android applications with hands-on exercises

Audience and prerequisites

This training is suitable for people with notions of offensive security but no prior experience in auditing Android applications. It is mainly aimed at pentesters and Android developers.

- Pentesters
- Android developers

Notions of offensive security and network and Unix knowledge are recommended.

Content

Day 1

Fundamentals: operation of an application and the Android ecosystem (services, intents, keystore, APK format, cache file, shared prefs, backup mechanism). Static analysis: analysis of permissions and interactions with the system and other applications, presentation of analysis tools and explanation of common artifacts giving information about the activities of an application.

Day 2

Dynamic analysis: architecture of an application at runtime, mechanism for interception and instrumentation of Java code, presentation of Frida and Objection to automate classic workarounds or obtain information. Practical cases: hands-on exercises on Android applications.

Password Cracking

1 days | Junior level



Description

Passwords still constitute an essential component of information system security today. During intrusions, different types of password hashes are recovered and being able to break them in a short time can prove decisive.

This training aims at presenting the techniques and tools for breaking password hashes as quickly as possible. A history of password storage developments will also be presented, to highlight bad examples and mistakes made in popular projects.

- 1 day (6 hours)
- Password cracking optimization techniques
- Datasets provided

Audience and prerequisites

This training is suitable for people having no prior knowledge of password cracking. It is mainly aimed at pentesters, system administrators, and developers.

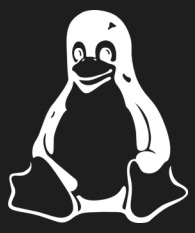
- Pentesters
- System administrators
- Developers

Content

Password storage and generation theory: storage type, hash functions, function attacks, candidate generation, computational technologies. History of algorithms. Series of practical exercises: identification of algorithms in source code, getting started with John the Ripper (candidate generation modes, development of derivation rules and candidate filters based on a password policy, dynamic formats, implementation or modification of a native format), getting started with Hashcat (advanced candidate generation with prince combination, siga genetic mutations and rule generation).

Embedded Linux Exploitation

5 days | Junior level



Description

Working with a real-world device, participants will progress through the complete vulnerability research workflow, from hardware analysis to exploitation. The course is designed for security professionals, embedded developers, and researchers interested in the offensive security of embedded Linux systems.

Objectives

- Identify critical hardware components and interfaces on an embedded Linux device.
- Access a device via UART, interrupt the boot process, and extract firmware.
- Analyze and emulate firmware components using QEMU and containerization.
- Perform static analysis and assist bug-finding with scripts.
- Set up and run a coverage-guided fuzzing campaign with AFL++ against a binary-only target.
- Analyze and exploit common vulnerabilities like command injection and memory corruption.
- Implement persistence mechanisms on a compromised embedded system.

Audience and prerequisites

This training is suitable for people working in cybersecurity, embedded systems development, and interested in offensive security of Linux embedded systems. In order to gain the most from this training, trainees must be:

- Comfortable navigating the filesystem, managing processes, and using common command-line tools.
- Able to understand basic TCP/IP concepts, ports, and protocols like HTTP.
- Familiar with reading and writing simple scripts in Python and a basic understanding of C programming concepts.
- Familiar with basic knowledge of computer architecture, ARM assembly language, and experience with a disassembler is a plus but not strictly required.

Content

Day 1

Setup and Extraction: Open-source information research, attack surface definition and analysis of hardware components. First boot sequence understanding and first extraction of the firmware.

Day 2

Analysis and Emulation of the Firmware: Software emulation techniques, environment setup and target processes identification. Extraction and emulation of important software from the firmware.

Day 3

Static analysis and Exploitation: Static analysis with tools lie Ghidra and semgrep for differential analysis and vulnerability research. Exploitation methodology and writing of first script for command injection.

Day 4

Fuzzing: AFL++ fuzzing introduction. Entry corpus creation, harness development and fuzzing campaign launch on a binary target.

Day 5

Advanced exploitation and persistence: The fuzzer will generate crashes. The goal on day 5 is to analyze those crashes and identify an exploitable vulnerability. ROP chain creation, exploitation script writing and system persistence techniques study.

Android for Security Engineers

5 days | Intermediate level



Description

Android is one of the most popular mobile operating systems on the market. While originally based on Linux, it stands out with specific components making it unique and significantly different from the traditional OS. During this training, the participants will discover the architecture of Android and the interactions between its different internal components. The system allows third-party applications to run while protecting end-user data.

Key components of the system will be analyzed, including the boot process and security mechanisms. The trainers will detail the evolutions of the versions starting from Android 10 and will discuss certain particularities of the manufacturers. The concepts presented will be put into practice through concrete exercises.

At the end of this training, participants will have an in-depth understanding of Android and will be able to be autonomous in any research work on this ecosystem.

- 5 days (35 hours)
- 15h theoretical courses / 20h practical labs

Audience and prerequisites

This training is an advanced level course designed for security engineers wishing to conduct research on this system.

- Pentesters
- Android developers
- Security engineers

Good knowledge of C development as well as basic knowledge of Linux systems is recommended.

Content

Day 1

Overall architecture of Android, boot chain, update system, security model and rooting a smartphone.

Day 2

Format of applications (APK) and presentation of compilation and debugging tools (exercises with Frida).

Day 3

Android Runtime, IPC mechanism (Binder) and presentation of the Bionic library (Android libc).

Day 4

Application life cycle: installation, startup, execution and shutdown. Exploration of traces/logs that may be present on a device. Encryption of user data.

Day 5

Final exercise: modification of an Android environment via Magisk modules and putting into practice the concepts learned during the week. Analysis of the specifics of the Linux kernel for Android.

iOS for Security Engineers

5 days | Intermediate level



Description

iOS is one of the most popular operating systems on the market, offering a state-of-the-art security model. During this training, participants will discuss the ecosystem and the fundamental building blocks of the iOS operating system. They will discover how to use the macOS compilation chain to deploy a program, then debugging and diagnostic tools.

The fundamentals of reverse-engineering applications and system services will be covered in a second step: the internals of Objective-C, the IPC mechanisms (mach, XPC, NSXPC) and the kernel APIs. Practical examples and exercises will guide participants throughout the training. Finally, software and hardware security measures specific to iOS will be covered, both in the kernel and user space.

- 5 days (35 hours)
- 18h theoretical courses / 17h practical labs

Audience and prerequisites

iOS for Security Engineers is an intermediate level training course designed for security engineers wishing to carry out research on this system.

- Pentesters
- iOS developers
- Security engineer

Good knowledge of C development and basics in reverse engineering are recommended. An IDA Pro license with the Hex-Rays decompiler for ARM64 is a plus.

Content

Day 1

Introduction: presentation of the working environment, development on Apple platforms (iOS and macOS), use of diagnostic tools, introduction to the Apple ecosystem.

Day 2

Introduction to reverse engineering on Apple platforms: update extraction, important file formats and tools, discovery and experimentation with the inner workings of Objective-C, introduction to the XNU kernel.

Day 3

Mach mechanisms: explanations and exercises around the XNU IPC API, presentation and exercises on the implementation of the Mach API for interaction with kernel objects, use of Frida to instrument services.

Day 4

Reverse-engineering of Mach services: theory and practical exercises around XPC and NSXPC, the abstractions used for inter-process communications. Overview of the use of signed pointers on Apple platforms.

Day 5

XNU security: presentation of the MACF framework, explanations of how AMFI works and isolation policies (sandbox), description of XNU defense-in-depth mechanisms, hardware security countermeasures in the kernel, mitigations of kernel vulnerabilities. Case study on diagnostic data upload.

Advanced IDA

5 days | Advanced level



Description

Hex-Rays is one of the major players in the development of reverse engineering tools. Their IDA product has established itself over the years as the benchmark in this area. However, the lack of documentation and resources sometimes makes it difficult to use.

The objective of this training is to familiarize yourself with IDA (its interface, its functionalities, its API and its ecosystem) through several theoretical and practical modules. Participants will also learn how to develop scripts and plugins to extend the functionality of IDA and its decompiler.

- 5 days (35 hours)
- 8h theoretical courses / 27h practical labs

Audience and prerequisites

This advanced level training is designed for security researchers and reverse engineering experts wishing to change environments or improve their use of IDA.

- Security researchers
- Reverse-engineering experts

Good knowledge of assembler (x86-x64, ARM) as well as Python programming is strongly recommended. An IDA Pro license (not supplied) is mandatory.

Content

Day 1

Introduction to IDA: terminology, architecture and presentation of the tool.
Understanding the Python SDK and API: basics.

Day 2

Getting started with the features available via different exercises. Static analysis: disassembler, FLIRT, IDS, Type Info Library. Dynamic analysis: debugger, tracer and binary instrumentation.

Day 3

Advanced programming (part 1): detailed presentation of the SDK and practice through scripting to automate complex tasks.

Day 4

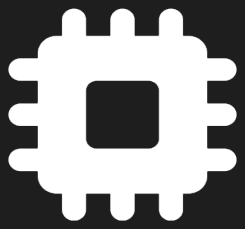
Advanced programming (part 2): development of plugins, loaders and processor extensions to practice on the previous notions.

Day 5

Extension of the decompiler: presentation of the Hex-Rays API, manipulation of microcode and AST, extension and improvement of the tools created during the session.

Hardware Intrusion

5 days | Intermediate level



Description

The objective of this training is to increase skills in hardware security analysis. It is aimed at both novices and those with an intermediate level.

At the end of this training, students must know the basic principles of electronics and soldering. They will be able to recognize the various components of a PCB and search for relevant information in component datasheets such as System on Chip (SoC) or external Flash to take advantage of them (RST implementation, debug functionality).

Finally, they will be able to identify possible test points, infer and then interact with the most common protocols (UART, JTAG/SWD, SDIO, SPI).

During the training, students will also learn to use equipment and tools useful for analysis (logic analyzers & Logic2, probes based on FT2232H & OpenOCD/flashrom)

- 5 days (35 hours)
- 17h theoretical courses / 18h practical labs

Audience and prerequisites

Hardware Intrusion Primer is a beginner to intermediate level training course designed for pentesters, security researchers and security teams.

- Pentesters
- Security researchers
- Security teams

Basic knowledge of electricity and electronics (how to use a multimeter, Ohm's law) is recommended.

Content

Day 1

Component Fundamentals: PCB, SoC, Flash, resistors, capacitors, transistors, crystal oscillators and PMIC.

Day 2

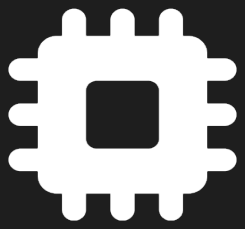
Theoretical reminders: electricity, security, analog and digital electronics.

Days 3 to 5

Common protocols: theory (characteristics, variation, usefulness in security analysis, signal shape) and practice (identify ports of interest, know how to use the hardware and tools to connect to them). Welding: principle, equipment and good practices.

DMA Attacks

4 days | Intermediate level



Description

Physical access to a machine opens up significant attack vectors, particularly through Direct Memory Access (DMA). These attacks allow an external hardware component to read and write directly to the target's RAM, bypassing the operating system, authentication, and traditional software protections.

During this four-day training course, participants will explore the hardware architecture of modern systems and learn how to leverage connectivity options (PCI Express, M.2, etc.) to perform memory dumps. The course covers the use of specific hardware tools (PCIScreamer, USB3380) and the PCILeech framework for injecting code or bypassing authentication on the fly. A significant portion of the training is also dedicated to reverse engineering for writing custom signatures, as well as forensic analysis of memory dumps using Volatility.

- 4 days (28 hours)
- Handling of specific interception equipment (PCIScreamer, USB3380)
- Balanced allocation of time between offensive exploitation (PCILeech) and memory analysis (Volatility)
- Practical exercises on Windows and Linux targets

Audience and prerequisites

This training is designed for technical professionals seeking to understand and exploit hardware vulnerabilities related to physical memory access.

- Pentesters / Red Teamers
- Security researchers and low-level developers
- Forensic and Incident Response (CSIRT) analysts

A solid understanding of operating system architecture (Windows/Linux) and the command line is required. Basic knowledge of reverse engineering (using IDA or Ghidra) and memory analysis is a significant advantage for the practical exercises.

Content

Day 1

Computer architecture: peripherals, data buses, key components, connectors, RAM types (frequencies, dual/triple channel). DMA concepts: history, access types, modern attack vectors, hotplug connectors. Security mechanisms: BIOS/OS level protections, security presence detection (IOMMU). Protection bypass: BIOS password erasure (clear CMOS, manufacturer passwords, brute force), IOMMU-related vulnerabilities. Capture hardware: presentation of the PCIScreamer and USB3380, connection methodology depending on the target, handling hardware issues (interference, compatibility). Labs: performing physical dumps of RAM via PCI Express, M.2, and USB3380.

Day 2

PCILeech: Internal workings and use of the tool. Exploitation: RAM extraction, memory manipulation, on-the-fly authentication bypass, kernel shellcode injection. Reverse engineering (Signatures): Presentation of Windows and Linux authentication mechanisms, identification of structures to modify. Analysis tools: Introduction to IDA/Ghidra for analyzing and patching these mechanisms. Labs: Writing a custom signature file (.sig) to successfully bypass authentication on a target Linux system.

Day 3

Post-exploitation techniques: dumping physical disks from administrator access (Linux/Windows), technical limitations. RAM analysis: introduction to the Volatility framework (v2 vs. v3), installation, internal workings, generating profiles/images for different operating systems. Reconnaissance: methods to determine the Linux version and type (via network, via a memory dump). Labs: performing a disk dump from Linux and Windows access, analyzing a Windows RAM dump with Volatility (tracking and searching for malicious processes).

Day 4

Volatility In-Depth: pedagogical refresher, review of the complex generation of image files (OS profiles). Investigation Labs: complete OS fingerprinting (network and memory), creation from scratch of a Volatility signature file for a specific Linux target, in-depth analysis of a Linux RAM dump to identify a persistent malicious process. Conclusion: summary of key points and attack vectors. Evolution of Defenses: points to monitor from vendors, deployment of DMA Guard, native integration and systematization of IOMMU in Windows.

Windows Forensics

5 days | Intermediate level

Description

Digital investigation makes it possible to reconstruct and understand in detail the chronology of a system's present and past activities. In the case of this training, we are interested in the Windows 10 and 11 operating system. Whether it is a security incident or a search for computer malware, the first responses aim to establish the perimeter of compromise and the attacker's methods. The technical approach presented is intended to be as exhaustive and reproducible as possible.

During these five days of training, the participants will be exposed to the forensic fundamentals in order to carry out a digital investigation for Windows and thus identify the traces of malicious intent. Each module will be illustrated by guided practical work allowing to apply the theoretical concepts previously taught. Finally, the training will conclude with a simulation of several traces (disk, memory, pcap).

This training is focused on the workstation and does not integrate the business dimension like Azure/AD (another training course will address this aspect soon).

- 5 days (35 hours)
- 11 course modules covering the fundamentals of Windows forensic investigation
- Cold or hot approach to cover several intervention situations
- Practical work on example artifacts

Audience and prerequisites

This training was designed for people with initial experience understanding Windows environments (administration, troubleshooting, advanced usage) and wishing to go further in the field of digital investigation. It requires basic knowledge of the Linux environment because this system is used to carry out some investigations.

- Advanced users (developers)
- System administrators
- Level 2 SOC analysts or from a cybersecurity team
- Beginner forensic analysts

Concepts of offensive security and good Windows & Unix knowledge are recommended to follow this training.

Content

Day 1

Getting started: training environment (virtual machine, Linux system). Reminders of the Linux command line. Windows: description of how Windows works (Windows history, processes, services, drivers, files, security model, network stack, main attacks). Windows events: description of the Windows logging model and the events to be aware of per use case. Scenario on event files.

Day 2

NTFS: study of the privileged file system of the Windows environment. MFT, USN Log and other special files. Decoding deleted dates and files. Reconstruct the chronology of events and pivot on an element (date, IOC). Registry: Registry contents. Use cases and configuration of the Windows system. Persistence Mechanisms: the means of persistence favored by an attacker are reviewed and thus identify the malicious programs executed by an attacker.

Day 3

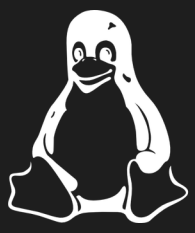
Execution of commands: traces linked to the execution of remote commands on the workstation through the different Windows protocols (WinRM, PsExec, WMI, RPC). Malicious code and files: analysis tools and methods allowing an initial study to be carried out on malicious code and thus extract the information of interest (behavior, IOC). By extension, files that can carry a malicious payload are also studied. Network protocols: particular attention is proposed in order to identify unusual network communications of a Windows system as well as the characterization of certain attacks (DNS tunnel, TOR).

Days 4 and 5

Artifacts: the study of most important forensic artifacts (prefetch, srum, amcache, navigation) in order to complete the timeline of the malware. Collection and acquisition methods are also presented in order to make the files available to be studied by the analyst (DFIR ORC). Memory analysis: techniques for acquiring and identifying suspicious elements are discussed to complement the analysis of offline elements. Running processes, network connections, cached files, memory injections and API hooking. Case study: several images are provided to the participants to put into practice all the techniques studied during the training. These images include various data such as a disk image, a memory capture and network captures.

Linux Forensics

5 days | Intermediate level



Description

Digital investigation makes it possible to reconstruct and understand in detail the chronology of a system's present and past activities. In this case, we are interested in the Linux kernel and two types of Linux distribution. While the examples and illustrations will focus apt and rpm-based distributions, most of the elements presented can be generalized to others.

During a security incident or a search for computer malware, the first questions deal with establishing the perimeter of compromise and the attacker's methods. The technical approach to such an investigation is intended to be as exhaustive as possible and, above all, reproducible.

During these five days of training, the participants will be exposed to the fundamentals in order to carry out a digital investigation for a Linux distribution and thus identify the traces of malicious intent. Each module will be illustrated by guided practical work allowing to apply the theoretical concepts previously taught. The training includes a role-play on several artifacts (disk, memory, pcap).

- 5 days (35 hours)
- 11 course modules covering the fundamentals of Linux forensic investigation
- Cold or hot approach to cover several intervention situations
- Practical work on example artifacts

Audience and prerequisites

This training was designed for people with initial experience understanding Linux environments (administration, troubleshooting, advanced usage) and wishing to go further in the field of digital investigation.

- Advanced users (developers)
- System administrators
- Level 2 SOC analysts or from a cybersecurity team
- Beginner forensic analysts

Concepts of offensive security and good Windows & Unix knowledge are recommended to follow this training.

Day 1

Getting started / the command line: training environment (virtual machine, Linux system). Reminder of the main commands for Linux. Linux and distribution: description of how Linux works including processes, file descriptors, security model (user/group, ACL, cgroup), named pipes, signals, terminal and command interpreter, X11. Filesystem: main filesystem types found in Linux systems (ext4, LVM, XFS). Specifics and special features for forensics: management of dates, deleted files, metadata, etc. Case of LUKS and virtual disks (qcow, vmdk).

Day 2

Boot sequence: identify the boot sequence in order to verify the integrity of the launch chain (grub, initramfs, UEFI case). Backdoor search on Systemd. Case of SecureBoot and kernel module signing. Program management: control of programs installed on the system (integrity, permissions). ELF format: program and library. Using apt and rpm package managers. Logging: type of logs (/var/log) and associated processes (syslog, auditd). Traces of compromise.

Day 3

Persistence mechanism: means of system and user persistence, device manager, Systemd. Process Analysis: process diagnostic tools, top Linux processes (ssh, X11), remote execution, procfs. Network Analysis: network configuration, network diagnostic tools, network socket, commonly encountered protocol and tunnel, network capture.

Day 4

Malicious code: analysis tools and methods allowing an initial study to be carried out on malicious code and thus extract the information of interest (behavior, IOC). Artifact: other artifacts (coredump, viminfo). Memory analysis: techniques for acquiring and identifying suspicious elements are discussed in order to complete the analysis of offline elements. Running processes, network connections, cached files, memory injections and API hooking.

Day 5

Container: Trace finding in containers and system reconstruction. Data collection: file extraction (disk copy) and selective (velociraptor). Case study: several images are provided to the participants to put into practice all the techniques studied during the training. These images include various data such as a disk image, a memory capture and network captures.

Mobile Forensics

5 days | Junior level



Description

The mobile phone has been evolving for several years as an extension of the workstation and is becoming a privileged target, because it is as close as possible to data. The digital investigation of this type of device aims to identify traces linked to criminal activities, to detect traces of malicious actions and compromise of the mobile phone.

This training aims at presenting the main artifacts present in the Android and iOS environments and to use an open source toolkit in order to analyze them. Adapted analysis methodologies will be presented in order to overcome the black box approach of certain systems and their pre-installed applications which complicate the audit of the phone.

This training exclusively addresses the case where the unlocking secrets of the phone are known.

- 5 days (35 hours)
- 2 mobile exploitation systems: Android & iOS

Audience and prerequisites

This training is suitable for people with knowledge of security or Linux system administration. It is mainly aimed at IT teams wishing to have first-level methods for investigating phones and who do not have software dedicated to this activity. More generally, anyone wishing to enrich their professional career with a security component in the mobile field.

- IT teams
- System administrators
- Security teams

Concepts of offensive security and good Unix knowledge are recommended to follow this training.

iPhone and Android phones are provided during the training for the hands-on exercises.

Day 1

Introduction and fundamentals: objectives, getting started, and an overview of mobile forensics (information sources, main formats of interest, timestamp management). iOS fundamentals: representation of the architecture and main services, security model, file systems, and specific data formats. iOS acquisition: acquisition methods illustrated by practical exercises in full acquisition (root access to the device) and partial acquisition (restricted access) using reference tools such as libimobiledevice, mvt, and sysdiagnosis generation.

Day 2

iOS system artifacts: review of the entire phone's activity to list execution traces and the presence of applications (info.plist, LSD database), analyze their activities (netusage, powerlogs, KnowledgeC, Biome) and their permissions (TCC). In-depth analysis of logs from archives using the sysdiagnose.py tool. iOS application artifacts: presentation of native and third-party applications and study of their common use cases (accounts, communications, web browsing). The practical exercises primarily rely on iLEAPP and Apollo to extract and analyze this specific data.

Day 3

Analysis methods: network capture, encrypted backup, and system diagnostics. Android fundamentals: architectures, OEMs, security model, file systems, with a detailed presentation of the ADB (Android Debug Bridge) and mvt-android tools. Android data acquisition: practical acquisition methods demonstrated through hands-on exercises covering device rooting, full data acquisition, and partial data acquisition targeting the collection of specific artifacts via ADB.

Day 4

Android System Artifacts: Review of phone activity and study of the main data formats (ABX, SQLite databases, Protobuf). Analysis of artifacts used to list applications, their permissions, and their activities, followed by a live analysis practical exercise using the data collected with MVT and ADB. Android Application Artifacts: Presentation of the classic directory structure of native and third-party applications, methods for retrieving and parsing their internal data. Study of use cases (accounts, communications, web) using ALEAPP as the central theme of the practical exercises.

Day 5

Malicious Android Applications: Mobile threat analysis and presentation of common injection methods. Static and dynamic analysis: Methodology for analyzing the structure of a suspicious APK using reverse engineering tools (apktool, mobsf, jadx) and dynamic execution within an emulator coupled with network captures. Practical exercises: Hands-on work dedicated to the manipulation and complete analysis of a known Android malware, including live analysis phases with ADB.

Kubernetes Forensics

3 days | Intermediate level



Description

Kubernetes has established itself as the orchestration solution for modern cloud-native infrastructures, managing the deployment and scaling of applications at large scale. This ubiquity now makes it an essential area of investigation for forensic analysts. The platform's distributed architecture, the variety of its components, and the volatile nature of its workloads significantly complicate the collection and preservation of digital artifacts.

In practice, Kubernetes is frequently used through managed services offered by the major cloud providers: AWS, Azure, and GCP. This training covers the specifics of these environments while exploring the inner workings of Kubernetes in a platform-agnostic manner, regardless of the underlying provider.

The training will first cover the fundamentals of containerization and the typical architecture of a Kubernetes cluster. Several forensic analysis approaches will then be presented: some rely on the underlying operating system of cluster nodes, while others leverage the logging mechanisms and native tools provided by cloud providers.

- 3 days (21 hours)
- Container and Kubernetes cluster analysis

Audience and prerequisites

This training is designed for individuals looking to discover Kubernetes forensics while deepening their knowledge of container analysis. It is intended for anyone involved in administering or investigating Kubernetes clusters.

- System administrators
- SOC / CERT analysts
- DevOps / DevSecOps engineers

A solid understanding of Linux and its shell environment is strongly recommended.

Content

Day 1

Introduction: getting started, and overview of Kubernetes investigation. Kubernetes and container threat landscape: volatility of information in Kubernetes, container internals, container runtimes. Fundamentals: cluster architecture, permissions model, Kubernetes weaknesses, exploration and identification of suspicious behavior.

Day 2

Collection and response: artifact collection checklist, CRIU and checkpoints, audit logs, and container isolation. Filesystem and memory analysis: overlayfs analysis, file recovery, memory analysis using checkpoints, image comparison.

Day 3

Network analysis and tooling: Kubernetes network model and forensic implications, traffic capture, DNS forensics and NetworkPolicies, syscall capture. Exercise: hands-on application of acquired knowledge through an incident response scenario under real-world conditions.

Malware Analysis

3 days | Intermediate level



Description

When dealing with security incidents, it is common to discover malicious code. This training aims to provide the keys to understanding malware and extract the elements of interest.

During the training, several types of malicious code are illustrated depending on the language used or the phase of the attack (exploitation, persistence). The different static and dynamic analysis methods are explained in order to provide complementary approaches to the analysis. A fairly significant part of the training deals hands-on exercises in the context of security incidents or operating procedures regularly observed in incidents. This course only addresses the case of userland malicious code.

- 3 days (21 hours)
- Malicious code analyzed on different languages
- Study of malicious files (Powershell, LNK, HTA)

Audience and prerequisites

This training is suitable for people who already have some experience in assembly (x86) or who have already undertaken program analysis (observing malware in sandbox). It is aimed at all people involved in handling malware, particularly security teams (SOC, CSIRT) or wishing to improve their skills on this subject.

- SOC analysts
- CSIRT analysts

Basic Windows/Linux knowledge is recommended to better understand how a malware works.

Content

Day 1

Qualification of a first level code: OSINT, automatic sandbox. Working environment: installation of an analysis environment (isolated/open) to process malicious code. PE/ELF structure: understand the format and aspects used by malware. Static and dynamic analysis of code: concepts and simple examples.

Day 2

x86(-64) assembler: first steps, control of execution flow and important instructions. Windows/Linux: suspicious APIs to know about and used by malicious codes. Disassembler 101: getting started, case of decompilers. Debugger 101: getting started, step-by-step study & breakpoint.

Day 3

Reverse engineering of real-world malware: PE malware, ELF malware, observation of interactions with the operating system. Malicious scripts: PowerShell deobfuscation and shellcode emulation. Analysis of concealment techniques: LNK, HTA.

Advanced Malware Analysis

3 days | Advanced level



Description

In today's evolving threat landscape, malware authors employ increasingly sophisticated obfuscation techniques to evade detection. This training provides hands-on training for security professionals seeking to sharpen their malware analysis skills.

Participants will use tools like Ghidra SRE, x96dbg debugger, and the MIASM framework to dissect complex malware targeting Windows and UNIX. Emphasis is on emerging threats like Golang malware, multi-stage payloads, BPF backdoors, and heavily obfuscated loaders. The training covers countering anti-analysis techniques (sandbox evasion, anti-debug), recognizing obfuscation/encryption patterns for automated deobfuscation via Ghidra scripting, recovering Golang symbols from obfuscated samples, and analyze BPF backdoors.

- 3 days (21 hours)
- Malicious code analyzed on different languages
- Study of multi-stage threats

Audience and prerequisites

This training is intended for people already having good knowledge of malware analysis but willing to strengthen their skill in reverse engineering of complex pieces of malware, and to extract useful information for incident response and threat hunting. It is mainly intended for incident response teams, and threat analysts.

Practical knowledge of malware analysis and knowledge of PE and ELF (x86-64) reverse engineering. Basic GNU/Linux and Windows internals knowledge.

Content

Day 1

Windows and Linux analysis disruption techniques: Checks and bypasses implemented by malware to dynamically protect their code and hide their behaviour. Strategies to reduce debugging detection and VM footprint.

Day 2

Overview of common obfuscation techniques: API Call encoding and encryption, Opaque predicates, Control Flow Flattening. Tools and Ghidra Scripting to recover symbols and reconstruct the execution flow. Multi-stage malware analysis: Analysis of complex malware implementing various strategies to evade detection, from the dropper to the final backdoor.

Day 3

Reverse engineering of threats: Golang and Rust malware analysis, understanding their respective structure and discovering dedicated obfuscation tools. Strategies to recover symbols and extract elements of interest.
BPF backdoor analysis: eBPF introduction, Case study of an advanced BPF backdoor.

Ransomware investigation

3 days | Intermediate level



Description

Ransomware attacks are the most feared threats to businesses. The urgency caused by the destruction, even partial, of the information system can quickly overwhelm your security teams if they are not prepared. This training allows you to better anticipate this type of incident and thus reduce the resolution time: reduce stress, anticipate the sequencing of activities to be carried out and quickly agree on the methods to be applied. The objective of this training is to provide the keys to investigate a ransomware incident whether it is localized to a small perimeter or generalized.

During the training, best practices for countering ransomware attacks will be provided, in particular in order to contain and stem the incident. Real race against time, training participants will thus be familiarized with the methods and tools to be implemented. It should be noted that only the technical investigation component and the first stages of the investigation are covered here.

- 3 days (21 hours)
- Identify the modus operandi of the main ransomware groups: practical approach
- Understand key step to start investigation and identify exfiltrated files

Audience and prerequisites

This training is suitable for individuals who have encountered security incidents before and are concerned about large-scale ransomware attacks. Technical skills are required in order to investigate this type of incident practically. The training uses the DFIR, ORC and Velociraptor tools to illustrate the process; prior knowledge of these tools or similar is advantageous.

- Internal cybersecurity team member in companies or agencies
- System administrator with cybersecurity skills
- Security manager wishing to understand the technical aspects

Good Windows knowledge is recommended to understand how attacks work (Active Directory, RDP, PowerShell, etc.). The manipulations are carried out on Linux environments.

Content

Day 1

Warning signs of an attack. First aid measures: preserving systems and backups, cutting off access and reducing the attacker's influence. Which traces to preserve first. Case of an attack by a service provider. Working in a compromised environment: myth, reality and pragmatism. Tools to use to share information and best practices for conducting investigations. Investigation during a crisis. Focus on the essentials to help with recovery. Challenging ecosystem. Legacy system & Edge device.

Presentation of DFIR ORC and Velociraptor.

Day 2

Ante-chronological approach. Quickly find the first structuring information, race against time. Identify malicious codes and backdoors. How to trace the attack. Case of administration positions / crisis positions. How to establish and share a situation (IOC / chronology / compromise perimeter). Analyst bias. Find the balance between exhaustiveness and efficiency.

Day 3

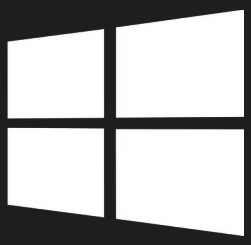
Approach by operating modes. Benefits and limits of CTI. Know the exposure surface and exploitation opportunities. Persistence mechanisms. OSINT to the rescue. The triad: VPN / phishing / vulnerable service. Effects sought by the attacker. Identify exfiltrated files and exfiltration mechanisms. Special case of false flag attacks, multiple attackers on a network.

Cyber Remediation: Best operational practices for crisis management and risk management. Regaining confidence in the information system and cleaning up its infrastructure.

Active Directory: Hardening &

Post-Compromise Recovery

5 days | Intermediate level



Description

For many organizations, Active Directory is the core of identity and access management. Its ubiquitous presence within information systems makes it a prime target for sophisticated cyberattacks and ransomware. Today, it is essential to make these infrastructures more resilient and to master the key post-intrusion steps to regain confidence in the system after an incident.

During this five-day training course, participants will acquire the skills necessary to defend and secure an Active Directory environment end-to-end. The program details the implementation of robust security mechanisms, the protection of directory control paths, and best practices in administration (Tiering model, LAPS, gMSA) to drastically reduce opportunities for compromise. Finally, a dedicated module prepares teams for the worst-case scenario: the presence of an attacker on the network and the methodology for regaining trust and cleaning the core of the information system.

- 5 days (35 hours)
- Full Active Directory environment simulating an enterprise infrastructure
- Defensive approach covering continuous hygiene, hardening, and post-incident response

Audience and prerequisites

This training is designed for technical professionals responsible for the design, security maintenance, and monitoring of Active Directory infrastructures.

- System and network administrators
- Security architects
- Members of security teams (Blue Team, SOC, CERT)

A solid understanding of Active Directory administration is required. Familiarity with PowerShell is a plus for confidently completing the practical auditing and configuration exercises.

Content

Day 1

Windows security architecture: security model, access token management, SID. Authentication mechanisms: NTLM vs. Kerberos in depth. Key concepts: how GPOs work, delegation. Anatomy of a compromise: Cyber Kill Chain (ransomware scenario), credential theft (LSASS dumping, SAM extraction), lateral movement techniques.

Day 2

Control paths: analysis and security (BloodHound/SharpHound), permission cleanup (ACL/ACE) on sensitive objects. Delegation of administration: principle of least privilege. Network services: securing DNS and DHCP, and disabling obsolete protocols (LLMNR/NetBIOS). Tiering model: concepts (Tier 0, 1, 2), deployment of administrative silos, Kerberos Armoring (FAST). Service accounts: implementation of Group Managed Service Accounts (gMSA) and protection against Kerberoasting.

Day 3

Administrative workstations: implementation of Privileged Access Workstations (PAWs). Privilege management: Just-in-Time (JIT) and Just-Enough Administration (JEA) administration, deployment of Local Administrator Password Solution (LAPS). Hybrid environments: cloud expansion (Entra ID), securing the Microsoft Entra Connect connector, understanding cloud-to-on-premises and on-premises-to-cloud attacks. Access control: implementation of Conditional Access and Multi-Facility Authentication (MFA) for critical roles.

Day 4

Visibility and Log Management: Advanced configuration of the Windows audit policy. Extended Monitoring: Deployment and configuration of Sysmon. Centralization: Analysis of critical logs and identification of essential Event IDs. Active Directory Hygiene: Auditing and compliance monitoring, regular verification of rights and privileges (PowerShell scripts), use of internal configuration analysis tools.

Day 5

Disaster scenario: Remediation Strategy methodology. Persistence analysis: tracking maintained access (WMI, scheduled tasks, backdoors). Recovery strategies: the concept of "AD failover" vs. "cleanup." AD cleanup: double password reset procedure (KRBTGT), attacker removal. Emergency hardening: creation and application of survival GPOs.

Data Breach: Investigations & Compliance

1 day | Junior level



Description

Data breaches represent one of the major risks for organizations, with often significant financial, reputational, and legal consequences. When a breach occurs, the speed and accuracy of the response are critical. It is essential to simultaneously implement technical remediation measures and legal and regulatory procedures.

During this one-day training course, participants will learn how to orchestrate a data breach response. The program adopts a dual approach: a technical component focused on quick wins and rapid investigation based on the type of threat (ransomware, web exposure, insider threats), coupled with a legal component detailing legal obligations, subcontractor management, and crisis communication. The objective is to provide immediately applicable operational reflexes.

- 1 day (7 hours)
- Dual approach: immediate technical actions (quick wins) and legal compliance
- Analysis of concrete investigation scenarios (ransomware, web vulnerabilities, misconfigurations)
- Focus on crisis communication and subcontractor management

Audience and prerequisites

This training is designed for technical and organizational professionals involved in cyber crisis management and data protection.

- Members of Incident Response Teams (CSIRT/SOC)
- Chief Information Security Officers (CISOs)
- Data Protection Officers (DPOs) and legal counsel
- Chief Information Officers (CIOs)

A general understanding of IT architectures and awareness of personal data protection issues (e.g., GDPR) are recommended. No advanced technical forensics skills are required.

Content

Context and initial detection: sources of detection (OSINT, dark web, internal monitoring, reports),

initial organizational responses. Leak assessment: evaluation of the impact, nature of the exposed data, volume, support for investigation and crisis scaling. Technical investigations (Quick Wins): scenario-based research and containment methodology; website (vulnerabilities, injections), ransomware attack (double extortion, exfiltration), insider threats (intentional leak, account compromise), misconfigurations (exposed cloud storage, open databases). Best practices: evidence preservation strategies, emergency hardening recommendations. Crisis communication: communication strategy surrounding the leak, key messaging, internal vs. external coordination, reputation management. Legal obligations and notifications: regulatory framework, legal deadlines for notifying supervisory authorities, informing affected individuals, incident documentation. Ecosystem and responsibilities: management of the specific case of subcontracting (contracts, chain of responsibility), understanding of the international context (cross-border transfers, extraterritorial laws).

Cloud Forensics in Azure

3 days | Junior level

Description

Microsoft Azure is widely deployed in many companies. Cloud services like M365 email, virtual machines, or managed databases are typical attack targets. As the keystone of these infrastructures' security, managing permissions and authentication play crucial role. Most scenarios encountered during incidents use cloud-native mechanism to compromise tenants.

Facing these threats, digital forensics evolve accordingly. Analyzing distributed logs, correlate events via APIs, and understanding the Azure ecosystem are indispensable. This course is designed to bring you the essential technical know-how and forensic methodology to analyze and respond to security incidents in the cloud.

- 3 days (21 hours)
- Microsoft Azure concepts, products and caveats
- Review of the most popular attacks and investigation procedures

Audience and prerequisites

This training is suitable for individuals who have encountered security incidents before and are concerned about incidents in the cloud. Technical skills are required in order to understand cloud attacks and connect to CLI console (from linux shell).

Previous usage and understanding of PowerShell is a plus.

Content

Day 1

Understanding Cloud Forensics: Key differences with traditional forensic. Setup an investigation environment: Essential tools and script to use. What is Azure? Explain what is Azure & M365. Where to activate and find logs. The License hell. EntraID: Sign-in and roles. Tenant configuration. Cloud Application: Understanding Conditional Access bypass. MS Graph API.

Day 2

M365: Business email compromise scenarios. Exchange online: Hunt for backdoored account. Azure subscription: Logs for resources. Virtual Machine, Azure storage and managed service. Tenant parameters. Logs mapping.

Day 3

Dive in KQL: Centralization strategy, query and explore, explore logs analytics. Checklist with investigation tools. Hands-on lab.

Cloud Forensics in AWS

3 days | Junior level



Description

Amazon Web Services (AWS) is the platform of choice for countless companies, making its services prime targets for attackers. Core components like EC2 instances, S3 buckets, and RDS databases are constantly under threat. The cornerstone of securing these infrastructures is Identity and Access Management (IAM), which governs all permissions and authentication. Most incident scenarios in AWS leverage cloud-native methods that are fundamentally different from those in on-premise systems.

Facing these threats, digital forensics must evolve. Analyzing distributed logs from services like CloudTrail and CloudWatch, event correlation via the AWS API, and a deep understanding of the AWS ecosystem are a must-have. This course is designed to provide you with the essential technical know-how and investigation methodology to analyze and respond to security incidents in the AWS cloud.

- 3 days (21 hours)
- AWS concepts, products and caveats
- Review of the most popular attacks and investigation procedures

Audience and prerequisites

This training is suitable for individuals who have encountered security incidents before and are concerned about incidents in the cloud. Technical skills are required in order to understand cloud attacks and connect to CLI console (from linux shell).

Previous understanding and usage of linux scripting is a plus.

Content

Day 1

Understanding Cloud Forensics. Key difference with traditional forensic. Setup an investigation environment. Essential tools and script to use. What is AWS ? Explain main core services. Focus on IAM. Understanding the logging system CloudTrail / CloudWatch.

Day 2

More on AWS logs. Investigate compromised EC2. Amazon AMI. S3. Common attacks. GuardDuty. Network logs and VPC.

Day 3

IR playbooks. Investigate large volume and create timeline. Amazon Athena. Help the analysis with third party tools.

Agentic AI Red Teaming

5 days | Intermediate level



Description

The development of agentic architectures marks a major technological breakthrough, transforming passive LLMs into proactive systems capable of orchestrating complex workflows where traditional algorithms reach their limits. This evolution enables the design of autonomous systems capable of reasoning, interfacing with third-party tools and databases, while maintaining a controlled and secure execution framework.

This five-day training course aims to impart the skills necessary to design these next-generation agents. The instruction is structured around a practical focus: the complete development of a Red Team agent dedicated to vulnerability recognition and identification. Participants will learn to manipulate state-of-the-art models, whether remote (Anthropic, OpenAI) or local (Ollama), and to integrate them into robust multi-agent architectures.

- 5 days (35 hours)
- Development of an agent for reconnaissance and vulnerability detection
- Advanced use of the Model Context Protocol (MCP) and smolagents
- Python development environment and access to provided models

Audience and prerequisites

This training is designed for technical professionals looking to develop their skills in AI-based autonomous systems engineering. It requires a strong familiarity with modern development environments.

- Software developers (intermediate to experienced)
- Penetration testers and security researchers
- SecDevOps engineers

A solid background in Python programming is essential. Participants should be familiar with the Linux command line and have a basic understanding of networking (TCP/IP, HTTP). While an interest in the LLM ecosystem is a plus, no prior AI knowledge is required.

Content

Day 1

Introduction to agent development: Key concepts of inference, state of the art of models (API vs. local), hardware sizing, and overview of frameworks. Scripting of queries to inference engines (online/offline) and programming of a first simple agent.

Day 2

Agentic Architecture and Tool Use: Thought-Action-Observation loop and stopping conditions. Function calls and typing. Agent architecture and specialization. Development of custom Python tools (tool calling). Implementation of a multi-agent architecture with orchestrator, worker, and analyzer. Implementation of CodeAgent capable of executing code in a segregated manner.

Day 3

Model Context Protocol (MCP): Introduction to MCP. Components: hosts, clients, and servers. The protocol, its transport layers (JSON-RPC), functionalities (tools, resources, prompts), network messages, and security (authentication, isolation). Presentation of python-mcp and fastMCP. Programmatically listing the tools, resources, and prompts exposed by an MCP server. Creating an MCP server exposing a network scanning tool. Creating an MCP server exposing dynamic resources (logs). Integrating an MCP client and using MCP with the agent.

Day 4

Memory, RAG, and Context Injection: Concept of a context window, limitations, and optimizations (short memory). RAG: Augmented Generation by Retrieval, theory, and libraries. Persistent memory via classic data structures. Implementation of a database (CVE) with VectorDB. Addition of a "Sumarizer" agent to optimize the context window. Implementation of a "remember" tool with persistent storage.

Day 5

Qualification, observability, production deployment, and security: Focus on the numerous security risks associated with agent deployment and best practices (sandboxing, proxying, caching, hardening). Presentation of monitoring and introspection tools. Presentation of testing strategies to validate workflows despite the inherent non-determinism of LLMs. Implementation of functional test scenarios and performance evaluation. Tool isolation and security enhancement. Agent finalization.

Network Interception in Rust

3 days | Intermediate level



Description

Mastering network traffic interception and manipulation is a critical skill for security analysis, penetration testing, and offensive tool development. This course offers a hands-on technical immersion, building a high-performance Man-In-The-Middle (MITM) tool from scratch using the Rust programming language.

The approach is entirely practical: participants design an interceptor capable of performing ARP poisoning attacks and processing complex application traffic. Through progressive workshops, the course explores the fundamental layers of the OSI model, frame decoding, and the implementation of transparent interception mechanisms for HTTP and TLS protocols. The use of Rust ensures safe memory management and optimal performance for low-level data processing.

- 3 days (21 hours)
- Offensive security and Rust development
- Development of a complete and functional MITM tool
- In-depth understanding of the network mechanisms being exploited

Audience and prerequisites

This training is designed for technical professionals who want to combine modern system development skills with offensive network expertise.

- Rust developers wishing to apply their skills to low-level security
- Penetration testers/auditors wanting to design their own interception tools
- Security engineers specializing in protocol analysis

A solid understanding of Rust fundamentals (ownership, structures, dependency management, error handling) is essential. Basic networking knowledge (OSI model, encapsulation, TCP/IP) is required to follow the technical modules.

Content

Day 1

Network and LAN layers: presentation of the layers manipulated in a MITM interception, analysis of local network operation, ARP resolution, and machine behavior in a LAN. Low-level development: development in Rust of the first transparent interception building blocks at the Ethernet, IP, and TCP layers

Day 2

HTTP and HTTPS: detailed presentation of the exchange process (structure, headers, and session mechanisms), implementation of HTTP interception (stream reconstruction, content inspection and modification). TLS interception: introduction to TLS mechanisms (handshake, certificates, constraints, and limitations), development of TLS interception. Validation: testing under real-world conditions in a physical environment simulating a local network, with complete end-to-end testing.

Day 3

Advanced protocols: a series of in-depth tests on the IP and TCP layers (option management, MSS, fragmentation, retransmissions), development and analysis of a second local network attack (DHCP spoofing). Performance and constraints: use of benchmarking tools in Rust (performance analysis, processing time measurement, optimization), consideration of real-world constraints (stability, robustness, scalability, detection by defense systems). Finalization: consolidation of the product code and final packaging of the MITM tool.

Advanced Rust

5 days | Advanced level



Description

The Rust language is now essential, offering developers the productivity of a modern, high-level language combined with strict robustness guarantees and a powerful type system. This training course aims to address the advanced concepts faced by Rust developers, enabling them to design complex, reliable, and highly maintainable software.

The topics covered, drawn from real-world experience, place a strong emphasis on practical application. Throughout the week, participants will develop a comprehensive multi-threaded data analysis and extraction tool. This project will be iteratively improved to incorporate dynamic search criteria, concurrent performance, an asynchronous client/server network interface, end-to-end testing, and control via a web interface.

- 5 days (35 hours)
- File scanner project, iteratively improved with each practical session
- Implementation of complex concepts: multi-threading, asynchronicity, and advanced serialization

Audience and prerequisites

This training is designed for technical professionals looking to enhance their Rust development skills.

- Software developers
- System and embedded engineers
- Code auditors and security researchers

Prior knowledge of Rust is essential: participants must be able to read Rust code and be familiar with its syntax and fundamental concepts (pattern matching, borrowing, etc.).

Experience in system or embedded development, and/or with C/C++, is a plus.

Content

Day 1

Error handling: approaches to managing fallible operations, using the Rust standard library or common crates, advantages and limitations of each approach, factors to consider, and maintainability. Panic: specific characteristics, explicit and implicit causes, idiomatic approaches to structuring code and reducing its occurrence. Optimal use of types, APIs, and idioms: common patterns (new type, builder), advantages and specificities of Rust typing for constraining program state or interface, functional APIs (combiners) for iterators/Option/Result, maintainability, performance of generated code, antipatterns, and common errors.

Day 2

Performance & multithreading: primitives and architectures of multithreaded processing, associated APIs in the standard library and in commonly used crates, tools for evaluating program performance (benchmarking and profiling), and identifying areas for improvement. These concepts are applied to the project through file scanner optimization, migrating the architecture to multithreading, and integrating statistical counters.

Day 3

Serde: concrete examples of Serde usage, configuration, and modeling to maximize its benefits. async/await: presentation of the async model, its advantages, disadvantages, specific features, and common pitfalls. Immediate practical application through the implementation of a network command/response protocol based on Serde and the development of asynchronous client/server interfaces.

Day 4

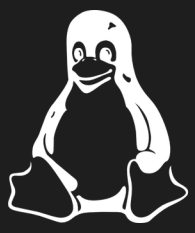
Testing: automated testing, structuring within a complex project, possible test levels (unit, integration, performance), tips for facilitating maintenance and efficiency, advanced tools (coverage measurement, snapshot testing). In-depth implementation of comprehensive integration tests validating end-to-end interactions between the scanner client and server.

Day 5

Web Development: concepts and integration of web development in Rust, applied to scanning control via the creation of a dedicated interface. Other topics (depending on time and participants' needs): conditional compilation and code build time (simple macros, procedural macros, build.rs), management, maintenance and auditing of a dependency tree, use of FFI to interact with C code, advanced use of the Clippy linter, combinator parsers and their advantages over grammar parsers (put into practice via MIME type filtering).

Linux Hardening

4 days | Intermediate level



Description

Linux systems form the basis of most infrastructures and workstations. However, a default configuration generally does not limit the impact in the event of a service compromise, nor does it hinder lateral movement. Securing these environments relies on mastering the OS's native restriction and isolation tools.

During this four-day training course, participants will cover four technical modules dedicated to hardening Linux systems: mandatory access control (AppArmor), network filtering (nftables), process containment (systemd), and rootless containerization (podman). These concepts will be systematically put into practice through dedicated labs, ranging from writing strict profiles to manually isolating processes using kernel primitives.

- 4 days (28 hours)
- 4 technical modules dedicated to restriction and isolation mechanisms under Linux
- Individual Linux virtual machines
- Strongly hands-on approach (writing AppArmor profiles, configuring nftables, systemd hardening, manual containerization and containerization via Podman)

Audience and prerequisites

This training is designed for technical professionals who want to design, administer, or audit hardened Linux systems.

- Linux System Administrators
- Secure Application Developers
- Security Engineers / DevSecOps

A solid understanding of Linux system administration (command line, process management, file systems) as well as basic networking (TCP/IP) and firewall (iptables/nftables) knowledge is required to complete the various modules.

Content

Day 1: AppArmor

Sandboxing mechanisms: overview of solutions under Linux, comparison of SELinux and AppArmor. AppArmor fundamentals: understanding a profile, cheat sheet of useful commands, common pitfalls. Creation and modification: modifying existing profiles, writing profiles from scratch. Advanced syntax: fine-grained control of transitions (clean exec, profile stacking), network access control, resource limits. Multiple launches: running two instances of a program with separate profiles. Labs: writing profiles, creating a jailed shell (read-only command prompt without network access).

Day 2: Nftables

Netfilter: the hook system, comparison of iptables vs. nftables. Nftables syntax: rule structure, conntrack and NAT management. Advanced filtering: filtering outbound traffic by UID/GID. Exploitation: counters, logging, and monitoring for rule debugging. Labs: setting up a state-of-the-art nftables firewall on a Linux VM, implementing outbound traffic filtering.

Day 3: Systemd

Overview: systemd architecture and project. Init management: replacing initrc, systemd services. Hardening: hardening systemd units (file system restrictions, network, capabilities). Advanced features: socket activation. System replacements: replacing cron (systemd timers), ifupdown (systemd-networkd), grub (systemd-boot), resolvconf (systemd-resolved), rsyslog (systemd-journald). Labs: auditing and hardening systemd units.

Day 4: Podman

Isolation concepts: an introduction to the basic tools of modern containers. Manual isolation (labs): isolating a process from the filesystem (chroot), isolating it from the network (unshare), controlling network access. Modern containerization: fundamental differences between Docker and Podman. Using Podman (labs): compiling exotic code (cross-compilation) without polluting the host system, launching an untrusted graphical application in isolation.

SYNACKTIV

